

Hierarchical Concept Bottleneck Models for Explainable Fine-Grained Bird Classification

Paniz Falahatinezhad (S337141) Erfan Moghadasian (S337136) Sina Ghiabi (S339390)

Explainable and Trustworthy AI – Politecnico di Torino, A.Y. 2025/2026

Reference teachers: Eliana Pastor, Eleonora Poeta

Abstract

The goal of this project is *explainability-by-design* for fine-grained image classification: instead of trusting a black-box CNN/Transformer, we want every prediction to be readable as a sequence of human-understandable visual concepts. We take a standard image backbone – in our case **ResNet-50** or **ViT-B/16** – and, instead of feeding its penultimate feature vector directly into a 200-way classifier, we *interpose two additional concept layers* between the feature vector and the final decision. The first added layer maps the feature vector to **13 coarse body-part probabilities** (e.g. bill, wing, tail, belly); the second layer maps the same feature vector to **312 fine attributes** (e.g. *bill colour: rufous*, *wing pattern: striped*). The two layers are explicitly coupled by a multiplicative *parent-conditioned gate*: each of the 312 fine attributes is gated by the probability of its parent body part, identically at training and inference, so the hierarchy is enforced architecturally rather than by a soft penalty. The final classifier is a single linear layer $W : \mathbb{R}^{312} \rightarrow \mathbb{R}^{200}$ that consumes *only* these 312 gated concepts. Every species prediction is therefore forced to be a transparent function of the bottleneck: for any image we can read off which body parts the model detected, which fine attributes were activated inside each part, and exactly which attributes pushed the predicted class. We use both ResNet-50 and ViT-B/16 in parallel to show that this explainable design transfers across very different feature extractors – the same kind of explanation comes out on top of either backbone – and we evaluate the resulting explanations on CUB-200-2011 along two axes: *concept quality* (per-attribute AUC – are the concepts correctly detected?) and *concept necessity* (per-part ablation – does removing a concept group actually change the decision?).

1 Goal – explainability by design

A standard image classifier $\hat{y} = \arg\max Wf$ feeds the penultimate feature vector f of a CNN or Transformer directly into a softmax layer. The vector f has no human meaning – there is no way to read *why* a particular species was chosen. The goal of this project is to remove that opacity at the architectural level: instead of letting f go straight into the classifier, we route it through two extra layers that are themselves human-understandable concepts, and force the linear classifier to consume *only those concepts*. The resulting model $\hat{y} = \arg\max W p_f$ is exactly as expressive as the classifier we trust the linear bottleneck to allow, and exactly as explainable as the 312 concepts it now consumes.

2 Hierarchical architecture

The pipeline has four stages, organised strictly top-down.

2.1 Stage 1 – backbone (feature extractor)

We use a standard image backbone in its frozen, off-the-shelf form: either ResNet-50 (336 px input, ImageNet-V2 weights) or ViT-B/16 (384 px, SWAG-E2E weights). The backbone produces a single dense feature vector f (2048-d for ResNet-50, 768-d for ViT-B/16). f is *not* the explanation – it is just the raw representation that the two added layers will translate into concepts.

2.2 Stage 2 – first added layer (13 body parts)

A small head W_c projects f to 13 probabilities, one per body part:

$$p_c = \sigma(W_c f + b_c), \quad p_c \in [0, 1]^{13},$$

with parts $\{\text{back, belly, bill, breast, crown, eye, forehead, head, leg, nape, tail, throat, wing}\}$. Reading p_c already answers “which body parts is the model looking at?”.

2.3 Stage 3 – second added layer (312 attributes with parent gate)

A second head W_f projects $[f; p_c]$ to 312 raw fine logits, gated by their parent body part:

$$p_f[i] = \sigma(z_f[i]) \cdot (0.5 + 0.5 p_c[\text{parent}(i)]). \quad (1)$$

Eq. (1) enforces the hierarchy architecturally: low parent probability damps every child attribute (floor 0.5, never zero, so gradients survive); high parent lets the child fire fully. The 34 global attributes (e.g. **shape**, **size**) use $\text{mask} = 1$. The gate is *identical* at training and inference, so the classifier never sees out-of-distribution p_f values.

2.4 Stage 4 – linear classifier on the bottleneck

$$\hat{y} = \arg\max (W p_f + b), \quad W \in \mathbb{R}^{200 \times 312}.$$

There is no residual side-channel from f to the classifier; the 312 gated concepts are a *sufficient statistic* for the prediction.

3 Training

The 13/312 hierarchy is parsed deterministically from the CUB attribute names; no extra annotation is re-

quired. Part labels are the OR-aggregation of their children. The model is trained in three sequential phases: phase 1 trains both heads on a frozen backbone with a loss $\lambda_c L_{\text{coarse}} + \lambda_f L_{\text{fine}} + 0.1 L_{\text{task}}$ (weighted BCE / Focal Loss / CE-LS); phase 2 calibrates *only* the linear classifier on the model’s own *predicted* p_f (so the head learns the actual distribution it will see at evaluation, not a clean GT distribution); phase 3 jointly fine-tunes everything with concept-dominant weights (2/2/1) and MixUp $\alpha=0.2$. The backbone uses an order-of-magnitude lower learning rate than the heads, and for ViT-SWAG the backbone rate is reduced further to preserve attention features.

4 Reading a single prediction

Because the classifier is a single linear layer over 312 concepts, every prediction can be reduced to a fully explicit causal chain (Fig. 1): the backbone produces f ; the first added layer turns f into p_c (which parts are visible); the second added layer turns f and p_c into the gated p_f (which fine attributes are present); and the predicted class is the one whose row in W aligns best with that p_f . To explain a specific prediction we simply rank the contributions $W[\hat{y}, i] \cdot p_f[i]$ for $i = 1 \dots 312$ and read off the top contributors. There is no probing, no surrogate model, no post-hoc attribution: the explanation *is* the decision.

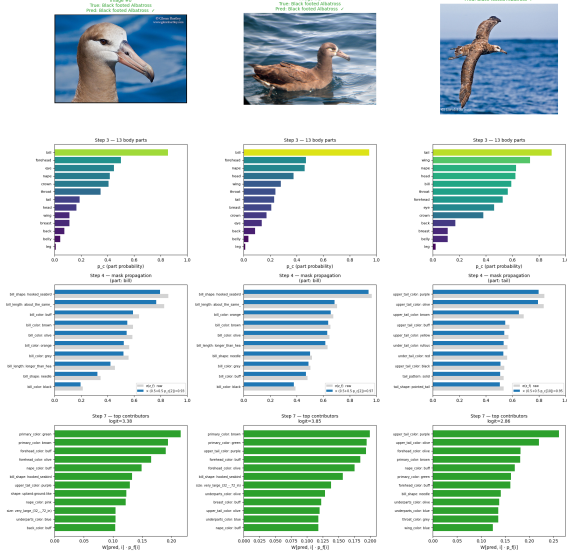


Figure 1: Reading a single prediction. From top to bottom: input image; 13 body-part probabilities p_c ; fine attributes inside the most active part, gated through Eq. (1); top contributions $W[\hat{y}, i] \cdot p_f[i]$ to the predicted class. The whole decision is visible on this single figure.

5 How well are the 312 concepts learned?

The explanation in Section 4 is only trustworthy if the 312 attribute detectors themselves are accurate. For each attribute we therefore compute its AUC against the CUB ground-truth attribute labels on the test split, and look at the 10 best- and 10 worst-detected attributes

for each backbone.

5.1 ResNet-50 instantiation

Fig. 2 shows the AUC ranking for the ResNet-50 H-CBM. The best-detected attributes (left) are visually obvious yellow body-part colours – both because yellow is salient and because it is well represented in CUB – and reach $\text{AUC} \geq 0.87$. The worst-detected attributes (right) gather rare colours (pink/purple body parts), the rare colour rufous and unusual shapes (owl-like, very_large); their AUC sits in the 0.36–0.49 band, i.e. barely above chance. The take-away for the explanation is direct: a prediction whose top contributors are yellow body colours is well-supported by the bottleneck, while a prediction that hinges on a rare colour or shape attribute should be read with caution.

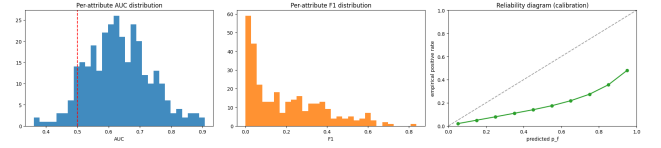


Figure 2: Per-attribute AUC of the ResNet-50 H-CBM on the test split. **Left:** the 10 attributes the model detects most reliably (yellow body-part colours, $\text{AUC} \geq 0.87$). **Right:** the 10 attributes the model detects worst (rare pink/purple colours, rare shapes, $\text{AUC} \leq 0.49$). Higher AUC means the predicted attribute probability genuinely ranks images that have the attribute above images that do not.

5.2 ViT-B/16 instantiation

Fig. 3 shows the same analysis for the ViT-B/16 H-CBM. The best-detected concepts are again the yellow body-part colours; the worst-detected concepts are again concentrated on rare colours. The structure of the explanation is therefore the *same shape* on both backbones, which is what we want from an explainable-by-design pipeline: the readable side of the model does not depend on which feature extractor was bolted under it.

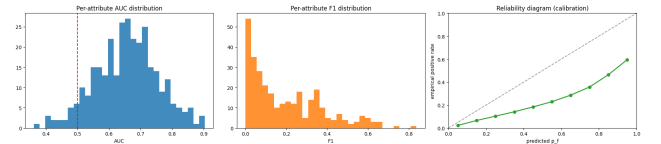


Figure 3: Per-attribute AUC of the ViT-B/16 H-CBM on the test split, same axes as Fig. 2. The set of easiest and hardest concepts is recovered *independently* by a completely different backbone, which means the explanation pattern is a property of the hierarchical bottleneck design rather than of a particular feature extractor.

5.3 Concept controllability

Faithfulness of the bottleneck can be tested directly: replace a fraction r of the 312 predicted concept probabilities with ground-truth values and re-run the linear classifier. At $r=0\%$ we get the normal baseline; at

$r=25\%$, 50% , 75% , and $r=100\%$ accuracy climbs monotonically toward the oracle (Fig. 4). There is no hidden residual side-channel – fixing more concepts always helps.

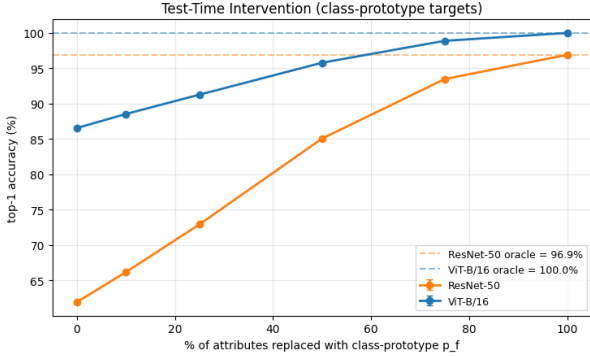


Figure 4: Concept-intervention curve for both backbones. At $r=0\%$ we recover the baseline (ResNet: 61.9%, ViT: 86.6%); as r grows accuracy climbs monotonically toward the oracle (ResNet: 96.9%, ViT: 100.0%). Both curves confirm that the linear classifier genuinely uses the concept layer.

5.4 Difficulty distribution across parts

Fig. 5 shows how the 312 attributes split into difficulty tiers (Easy: $AUC \geq 0.75$ on both backbones; Very hard: neither) across the 13 body parts. **tail** has the most Very-hard attributes ($n=14$) yet is also the most causally necessary – a clear direction for future work.

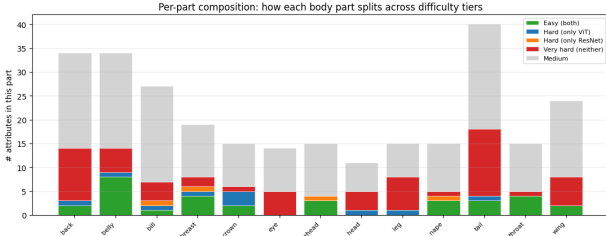


Figure 5: Per-part attribute difficulty composition. Each stacked bar is one body part; colours = AUC tier on both backbones jointly (green=Easy, red=Very-hard, grey=Medium). **tail** has the most Very-hard attributes despite being causally necessary.

6 Are the concept groups causally necessary?

A faithful explanation should not just be a list of concepts that *correlate* with the prediction – it should pinpoint the concepts the model genuinely *needs* to make the decision. We test this by zeroing the p_f columns of one body part at a time across the whole test split, re-running the linear classifier, and recording the resulting drop in accuracy. Because the classifier consumes p_f directly, this is a clean causal counterfactual: “what would the model do if it could not see this body part?”

6.1 ResNet-50 instantiation

Fig. 6 (left) shows the result for ResNet-50. The three body parts the model relies on most heavily are **bill**, **tail** and **belly**: zeroing **bill** alone drops accuracy by 1.4 pp, and the top three together account for more than half of the total ablation budget. The tail of the ranking (**crown**, **forehead**, **nape**) contributes less than 0.2 pp each.

6.2 ViT-B/16 instantiation

The same ablation on ViT-B/16 (Fig. 6, right) recovers the same three parts in the same order with very similar magnitudes. The *which parts matter* signal is a property of the H-CBM design, not of the backbone.

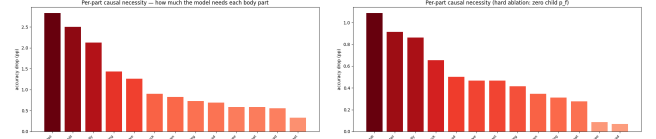


Figure 6: Per-part causal necessity: accuracy drop when child p_f columns of each body part are forced to zero. **Left:** ResNet-50. **Right:** ViT-B/16. **bill**, **tail**, **belly** dominate on both backbones.

7 Conclusion

We presented a hierarchical concept bottleneck model whose entire purpose is explainability: by inserting two human-understandable concept layers (13 body parts and 312 gated fine attributes) between any image backbone and the final linear classifier, we turn every prediction into a readable causal chain. We ran the same explainable design on top of two very different feature extractors (ResNet-50 and ViT-B/16); on both, the explanation surfaces the same way – the same yellow body colours come out as the easiest concepts, the same rare colours and shapes come out as the hardest, and the same three body parts (**bill**, **tail**, **belly**) come out as causally necessary. The H-CBM bottleneck is therefore an explainability layer that sits on top of *any* backbone, not a property of one particular architecture.

References

- [1] P.W. Koh *et al.*, “Concept Bottleneck Models,” *ICML*, 2020.
- [2] F. Pittino *et al.*, “Hierarchical CBM for Vision,” *EAAI*, 2023.
- [3] E. Poeta *et al.*, “Concept-based XAI: A Survey,” *ACM CSUR*, 2025.
- [4] T.-Y. Lin *et al.*, “Focal Loss for Dense Object Detection,” *ICCV*, 2017.
- [5] A. Kendall *et al.*, “Multi-Task Learning Using Uncertainty,” *CVPR*, 2018.
- [6] “P1 – Hierarchical Concept-Based Explainable-by-Design Models,” XAI Course brief, PoliTo, 2025/2026.